

HW: Local running of LLMs

Summary

In this HW, you are going to invoke LLMs locally - for chat and for code generation.

Q1. (6 points)

Install Ollama from <https://ollama.com/> - it lets you download and run LLMs locally! After it's installed, do this [to install Meta's llama3 model] - **ollama run llama3**:

```
(base) C:\Users\satyc>ollama run llama3
pulling manifest
pulling 6a0746a1ec1a... 1% ▢ 45 MB/4.7 GB 5.0 MB/s 15m17s
```

Similarly, get 'gemma2:2b' [do 'ollama run' or 'ollama pull' on it].

After both models are installed, you are ready to start sending prompts to them :)

pip install these: streamlit, ollama, litellm

Save the following as a .py (eg. invokeMultipleLLMs.py):

```
import streamlit as st
from litellm import completion

# set up the Streamlit app
st.title("Multi-LLM prompting")

# create a text input for user messages
user_input = st.text_input("Prompt:")

if st.button("Send"):
    if user_input:
        messages = [{"role": "user", "content": user_input}]

        # columns for side-by-side display
        col1, col2 = st.columns(2)

        # first
        with col1:
            st.subheader("llama3")
            try:
                response = completion(model="ollama/llama3", messages=messages)
                st.write(response.choices[0].message.content)
            except Exception as e:
                st.error(f"Error: {str(e)}")

        # second
        with col2:
            st.subheader("gemma 2:2b")
            try:
                response = completion(model="ollama/gemma2:2b", messages=messages)
                st.write(response.choices[0].message.content)
            except Exception as e:
                st.error(f"Error: {str(e)}")

        st.divider()

        st.write("Exercise for later - install two more models from https://ollama.com/libra")

        st.divider()
    else:
        st.warning("Please enter a prompt.")

# information about the app
st.sidebar.title("About")

st.sidebar.write("This app, derived from https://github.com/Shubhamsaboo/awesome-llm-apps, shows
```

```
st.sidebar.write(  
    "Aim - to see how different AI models respond to the same prompt..."  
)
```

Run the program, by doing this: '**streamlit run invokeMultipleLLMs.py**' - that should pop up your default browser that will display the streamlit app :) Now you can put in prompts that are sent to both LLMs, and look at their outputs side by side - cool!

Multi-LLM prompting

Prompt:

Omg!

Send

llama3

What's up?! It looks like you're excited about something! Would you like to share what's got you so pumped? I'm all ears and ready to chat!

gemma 2:2b

👋 What's going on? 😊

Did something exciting happen, or are you just feeling enthusiastic? 🍌 Let me know! 😊

Exercise - install two more models from <https://ollama.com/library>, add two more columns above, put in code to send your prompt to FOUR LLMs and display their responses.

Q2. (4 points)

Do 'ollama run qwen2.5-coder:7b' [or pick another model from <https://ollama.com/library/qwen2.5-coder> and install that] - that will download the model, then put up a console when you can enter text prompts that result in code generation! Eg:

[illegible]

After the course you can continue creating more local LLM apps including RAG, and, even create **agents** that run locally [eg to perform search, look up stock prices...] - start with <https://pypi.org/project/phidata>, <https://docs.phidata.com/examples/introduction>,

<https://serpapi.com>, and <https://docs.phidata.com/tools/serpapi>.

Getting help

Please post your qns (and answers!) on the 'hw5' Piazza page.

What to submit

3+2=5 screenshots, as a .zip file.

ENJOY! Hope this serves as a starting point for much more exploration :)
